

02YMECH - Solution 9

Assigned for the week of Nov 17, 2025

Solutions

1. Integrating the velocities,

$$\mathbf{r}_1(t) = (1 + t, \tfrac{1}{2}t^2), \quad \mathbf{r}_2(t) = (-1 - t, \tfrac{1}{2}t^2), \quad \mathbf{r}_3(t) = (0, \tfrac{1}{2}t^2).$$

- (a) Center of mass:

$$\mathbf{R}_{\text{CM}}(t) = \frac{1}{3} \sum_{i=1}^3 \mathbf{r}_i(t) = (0, \tfrac{1}{2}t^2).$$

Velocity and acceleration:

$$\mathbf{V}_{\text{CM}}(t) = (0, t), \quad \mathbf{A}_{\text{CM}} = (0, 1).$$

The center of mass does *not* move with constant velocity.

- (b) Total linear momentum:

$$\mathbf{P} = \sum m\mathbf{v}_i = (1 - 1 + 0, t + t + t) = (0, 3t).$$

Since $\dot{\mathbf{P}} = (0, 3) \neq 0$, linear momentum is *not conserved*.

- (c) Total angular momentum about the origin:

$$L_z = \sum m(x_i v_{iy} - y_i v_{ix}).$$

For each particle,

$$L_{1z} = t, \quad L_{2z} = t, \quad L_{3z} = 0,$$

so

$$L_z = 2t.$$

Since $dL_z/dt = 2 \neq 0$, angular momentum is *not conserved*.

2. (a) Total mass:

$$M = \int_0^a \int_0^b \sigma_0 \frac{x}{a} dy dx = \frac{\sigma_0 ab}{2}.$$

- (b) Center of mass:

$$x_{\text{cm}} = \frac{1}{M} \int_0^a \int_0^b x \sigma_0 \frac{x}{a} dy dx = \frac{2a}{3},$$
$$y_{\text{cm}} = \frac{1}{M} \int_0^a \int_0^b y \sigma_0 \frac{x}{a} dy dx = \frac{b}{2}.$$

Thus

$$(x_{\text{cm}}, y_{\text{cm}}) = \left(\frac{2a}{3}, \frac{b}{2} \right).$$